



LINE-FIT
FLIGHT DECK DOOR
SURVEILLANCE SYSTEM

(FDDSS)

AIRBUS A220

INTRODUCTION



REGULATIONS



ICAO



MATERIALS



INTRODUCTION



AIRBUS



Satisfies the requirement for a non-intrusive, non-distracting method of identification by the Captain and First Officer of who is trying to gain access to the Flight Deck.

Acts as deterrent to terrorism and air rage and helps to prevent catastrophic loss of aircraft from terrorism or revenge motivation.

Represents low-cost option to significantly increase levels of aircraft safety and security.

System forms part of the Airbus A220 Type Certificate and is certified wherever the A220 is certified to fly.





ICAO



REGULATIONS

Global regulators strongly recommend video systems...

Many European carriers proceeded with the installation of Flight Deck Entry Video Surveillance Systems after 9/11 and before mandates were issued.

ICAO Annex 6 – Part 1 – Chapter 13 – Standard 13.2.3(b) (1 Nov 2003):

"means shall be provided for monitoring from either pilot's station the entire door area outside the flight crew compartment to identify persons requesting entry and to detect suspicious behaviour or potential threat."

...interpreted into the Eurocae Minimum Operating Specification ED-123 which describes the minimum specification for a cockpit door surveillance system.



EU COMMISSION REGULATION (EC) No 859/2008 Sub-Part S OPS 1.1255 (20 Aug 2008):

"In all aeroplanes which are equipped with a flight crew compartment door...means shall be provided for monitoring from either pilot's station the area outside the flight crew compartment to the extent necessary to identify persons requesting entry to the flight crew compartment and to detect suspicious behaviour or potential threat."

MATERIALS



FV-0413 Camera



FV-0710 VTU



FV-0950 CCU



MATERIALS



FV-0413 CAMERA



AD Aerospace's FV-0413 is domed internal aerospace camera, including IR Illumination.

The Camera is designed to be fitted within the aircraft ceiling trim, in a pressurised, heated area.

It incorporates IR (Infra-Red) illumination operating at 860nm, outside the bandwidth of the human eye to provide superior low light performance.

The camera contains the very latest generation of analogue video sensing technology and offers excellent low light performance and facial recognition.

Specifications	
Size	Size: 100mm diameter including the flange x 80mm deep. Size: (Dome Protrusion) 60mm diameter x 40mm deep.
Weight	0.4kg Maximum
Material	The housing is currently manufactured as a machined part from a solid block. It consists of a painted aluminium housing, with nonadjustable fixed dimensions, and a flat aluminium lid. Fabricated Parts: Al Alloy to BSEN485:3103H4/5251H22, or 6061T4.
Fixings	The camera is secured using 3 fixing hole positions in the flange.
Cooling	There is no requirement for air cooling and the unit contains no internal fans.
Power Requirements	The power drawn is to be a maximum of 220mA from the 28V source.
IR Performance	IR Performance: 6 LEDs, at +/- 45 degrees cone, at 860nm Range >10ft.
Connectors	Mil-C-38999 connector
MTBF	50,000 hours.

MATERIALS

FV-0710 VIDEO TRANSMISSION UNIT



AD Aerospace's FV-0710 is a fully qualified (DO-160G) robust digital Video Transmission Unit (VTU) designed to be used in commercial aerospace applications.

It is capable of streaming digital video over IP Ethernet links from up to 8 100Ω balanced composite video signals.

The VTU is contained in a simple compact aluminium box housing and can be fitted in the overhead area on aircraft as well as any Avionics Bay or even behind cabin trim due to its compact size.

Video signals are multiplexed, digitised and compressed in the VTU, then stored within the on-board removable storage (CF Card) with maximum combined rate of up to 100 pictures per second (@ CIF).

A 64Gb device gives 26 hours of recording at 4 frames a second with 8 cameras, or 52 hours with 4 cameras.



Specifications	
Size	250mm x 110mm x 58mm (excluding mounting brackets)
Weight	1.2kg Maximum
Qualifications	RTCA Do178B and Do160D / E.
Input	Video: 8 Balanced composite video inputs. Audio: 2 Inputs
Output	Ethernet: 3 Outputs
Ethernet	10/100 base T to IEEE 802.3
Power Input	28V dc to RTCA Do160D
Power Consumption	42W maximum on start up 14W normal operating conditions
Other	• Electronic Flight Bag compatible • Removable Flash Storage

MATERIALS

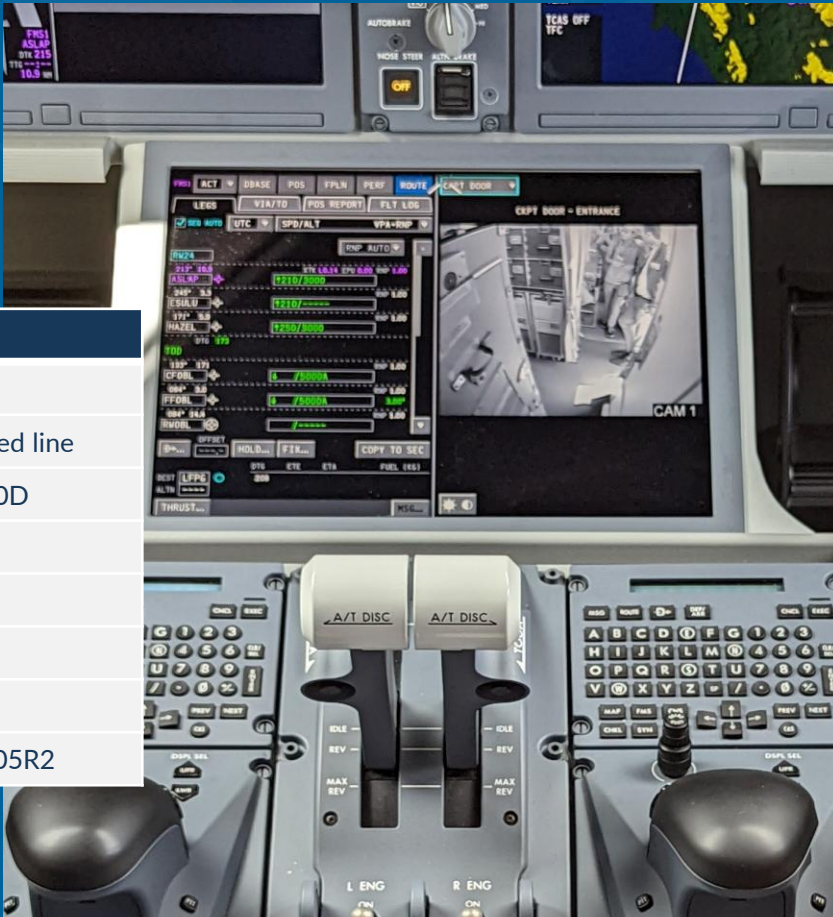
FV-0950 CAMERA CONTROL UNIT



The Camera Control Unit is used in this instance to interface between the monitor output of the FV-0710 VTU and the Airbus A220 MFD.

The Camera Control Units provide regulated, filtered DC power to the cameras, as well as providing a 100-ohm balanced video signal drive output for distances of up to 100 meters (328 ft), ensuring a cleaner image.

Specifications	
Video Inputs	1.0V p-p into 75 Ohms unbalanced
Video Output	2 off 1.0V p-p into 100 Ohm balanced line
Power Consumption	250mA @ 28V dc per RTCA DO-160D
Weight	240g
Qualification	RTCA DO-160D
MTBF	50,000 hrs
MTBUR	40,000 hrs
Boeing Qualified	D6-36440 through TSA BTS-03-7905R2



INSTALLATION



CONTACT US

Mark Isherwood (Sales Executive)
misherwood@ad-aero.com
<https://www.ad-aero.com/>
+44 (0)1928 280312

**ELECTRA HOUSE
PEPPER ROAD
HAZEL GROVE
STOCKPORT
SK7 5BW
UNITED KINGDOM**

